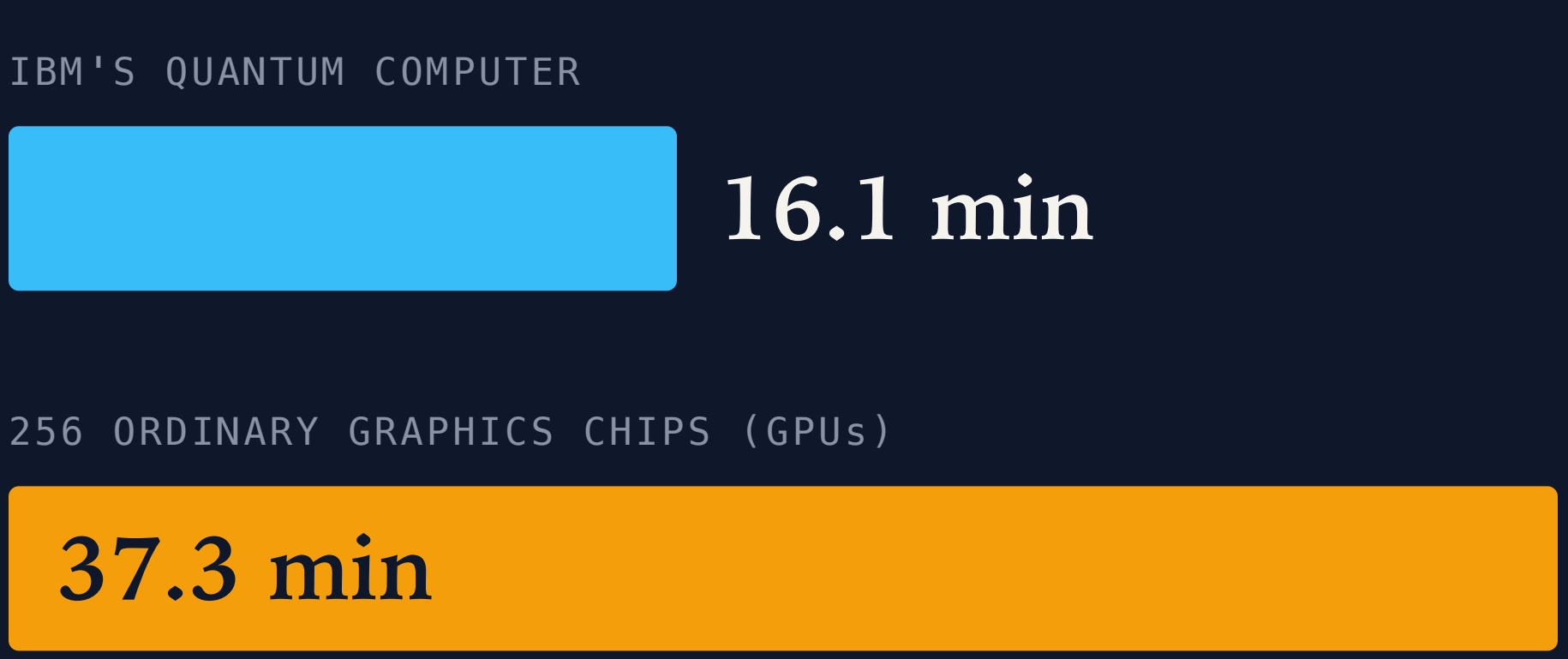
Quantum won by 21 minutes.

Then a team in Singapore matched it on ordinary chips, and IBM added them to the paper.



Same problem, same answers. Faster isn't the same as "impossible without quantum."

IBM + Algorithmiq, Jul 30 (arXiv 2607.25941, v3 adds the SUTD authors). SUTD reproduction: arXiv 2608.13110.

QUANTUM IN 60 SECONDS

Four words to know

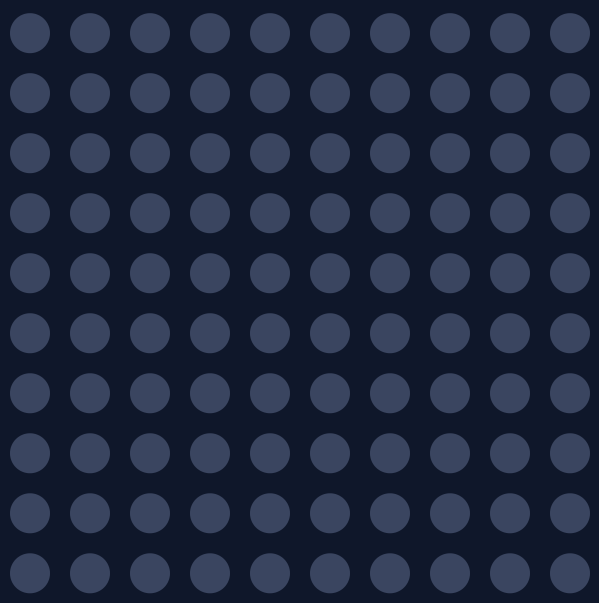
QUBIT	A quantum bit. Not a strict 0 or 1: it can hold a blend of both, and link with other qubits. Powerful, and very fragile.
LOGICAL QUBIT	Many fragile qubits bundled into one dependable qubit that catches its own errors.
FAULT-TOLERANT	A machine reliable enough to run long, useful jobs. The goal of the whole field. Nobody has one yet.
QUANTUM ADVANTAGE	A quantum computer doing something useful that no normal computer can match.

Heat, vibration or a stray signal scrambles a qubit. That's why everything below is about errors.

THE REAL STORY OF THE YEAR

How few fragile qubits can you get away with?

JAN · GOOGLE

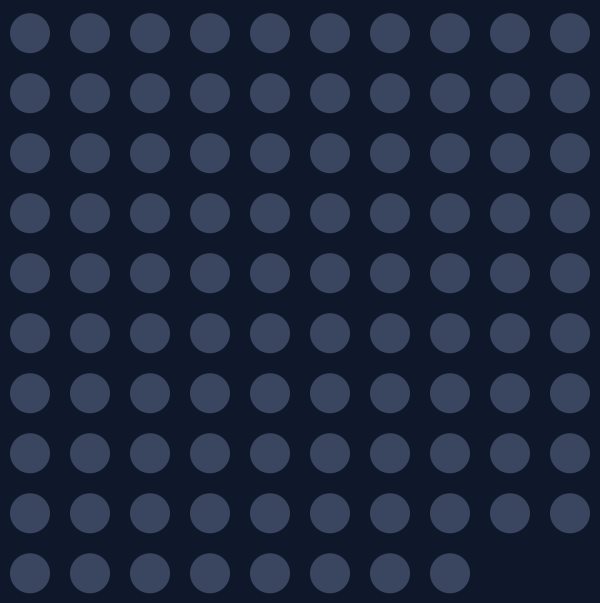


↓ bundled into

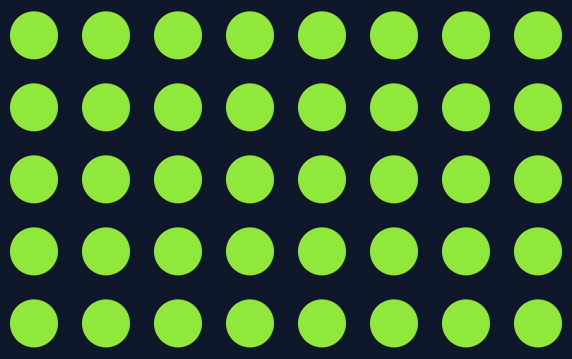


~100 → 1

MAR · QUANTINUUM



↓ bundled into



98 → 48

Grey = fragile (physical) qubits · green = dependable (logical). Different error-correction schemes, so not like-for-like; Google ratio approximate (Willow, 101 qubits).

DEVELOPMENTS · DATED

Quantum's 2026, dated

● science / hardware ● money

- **MAR 10** Quantinuum: 48 dependable qubits from 98
- **MAR 16** NVIDIA: graphics chips fix errors live
- **JUN 2** IBM commits $10B+ over five years
- **JUN 4** Quantinuum IPO raises $1.68B
- **JUL 3** IQM goes public
- **JUL 30** IBM claims "quantum advantage"
- **AUG** Singapore team matches it on GPUs
- **AUG 19** IBM links two ultra-cold fridges
- **SEP 8** US awards that take equity stakes
- **OCT 8** Tonight: IBM's Muir Kumph

Ahead: **Jan 1, 2027** US national-security buys need quantum-safe encryption · **2029** IBM & Microsoft fault-tolerant targets

Sources: Quantinuum; NVIDIA; IBM; CNBC; Commerce Dept.; arXiv. Full links in the first comment.

THE QUESTION I'M TAKING IN

What would convince you Kookaburra works?

IBM's first error-corrected building block, due this year. Not shown as of mid-September.

NYC Quantum Computing Meetup
Muir Kumph · IBM Research
IBM's path to fault tolerance
Thu Oct 8 · 6pm · 417 5th Ave
Next: what the room said · 2/2